MIL-S-22432/68
22 August 1974
SUPERSEDING
MIL-S-22432/6A
19 MARCH 1968
MIL-S-22432/6(USAF)
30 AUGUST 1963

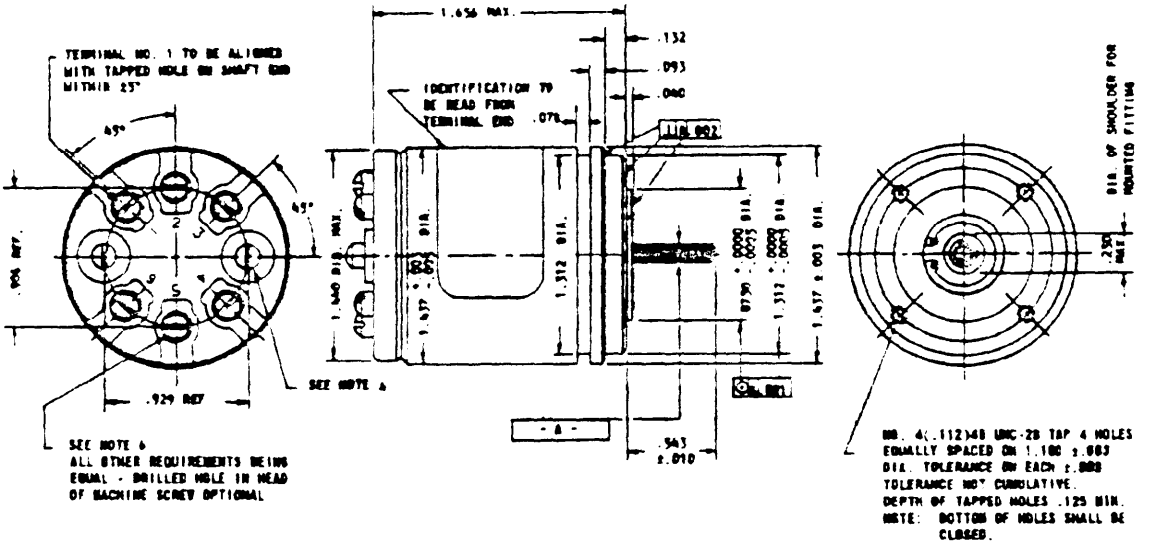
MILITARY SPECIFICATION SHEET
SERVOMOTOR, TYPE MARK 7 MOD 7

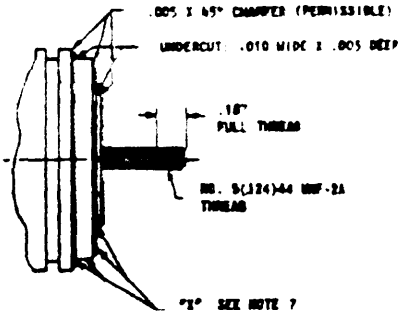
This Specification is mandatory for use by all Departments and Agencies of the Department of Defense.

THE COMPLETE REQUIREMENTS FOR PROCURING THE SERVOMOTOR DESCRIBED HEREIN SHALL CONSIST OF THIS DOCUMENT AND THE ISSUE IN EFFECT OF SPECIFICATION MIL-S-22432

SPUR GEAR DATA	
TEETH	13
PITCH	120
OUTSIDE DIAMETER	.1247 +.0000 -.0005
PRESSURE ANGLE	20°
TOOTH FORM	FULL DEPTH INVOLUTE
DESIGN TO AGMA 207.04 (ASA B6.7-1956) "20 DEGREE INVOLUTE FINE PITCH SYSTEM" PRECISION GRADE 1 TESTED TO AGMA 236.04 (ASA B6.11-1956) INSPECTION OF FINE GEARS.	

UNLESS OTHERWISE SPECIFIED
DIMENSIONS ARE IN INCHES
TOLERANCES ON FRACTIONS ±1/64
DECIMALS ±.005 ANGLES ±2°





NOTES:

1. PERPENDICULARITY AND CONCENTRICITY SHALL BE MEASURED WITH THE UNIT IN A VERTICAL POSITION SUPPORTED BY THE SHAFT. THEN THE HOUSING SHALL BE ROTATED.
2. RUNOUT OF FEATURE 'B' SHALL NOT EXCEED .0014 TIR.
3. END PLAY AND RADIAL PLAY SHALL BE MEASURED DURING THE REVERSAL OF THE FOLLOWING LOADS:
   END PLAY - 1 POUND RADIAL PLAY - 1/2 POUND
4. THE DESIGN OF THIS UNIT SHALL PROVIDE FOR THE TERMINAL BLOCK TO REMAIN FIXED WHEN TERMINAL SCREWS ARE REMOVED.
5. THE END CAP DESIGN SHALL BE SUCH THAT THE TERMINAL LUG MAY BE INSTALLED WITH THE BARREL INBOARD OR 180° POSITION OUTBOARD. RECESSES FOR LUGS SHALL BE OF WIDTH AND DEPTH TO PREVENT TURNING OF THE LUGS.
6. THE FOLLOWING LIST OF HARDWARE: MS-35276-213 MACHINE SCREW (8 REQUIRED) MS-35336-135 LOCK WASHER (8 REQUIRED) SHALL BE PACKAGED IN A VENTILATED ENVELOPE MADE OF MATERIAL HAVING A PH VALUE OF 6-8 THAT IS COMPATIBLE WITH THE WEIGHT OF THE CONTENTS. THEN THE PACKAGED HARDWARE SHALL BE PLACED IN THE UNIT SERVOMOTOR PACKAGE.
7. HOUSING SHALL BE STAINLESS STEEL. SURFACES MARKED 'X' SHALL BE BRIGHT METALLIC FINISH (AS PER MIL-C-13924).
8. DIMENSIONS AND TOLERANCES COMPLY WITH ANSI Y14.5.
9. PENDING ISSUANCE OF A QUALIFIED PRODUCTS LIST (QPL) A SAMPLE OF THE SERVOMOTORS TO BE FURNISHED UNDER THIS SPECIFICATION SHALL BE SUBJECTED TO THE PREPRODUCTION TESTS AS SPECIFIED IN SPECIFICATION MIL-S-22432.

FSC 6105

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TABLE I
SERVOMOTOR, TYPE MK 7 MOD 7

Performance Requirements and Characteristics

Requirements	Value	Unit	Tolerance
Electrical:			
Frequency:			
Rated	400	hz	
Test	400	hz	±1%
Range	380 to 420	hz	
Voltage:			
Reference phase, rated	115	volts rms	
Reference phase, test	115	volts rms	±1%
Reference phase, range	103.5 to 126.5	volts rms	
Control phase, rated (series)	115	volts rms	
Control phase, test (series)	115	volts rms	±1%
Control phase, range (series)	0 to 126.5	volts rms	
Control phase, rated (parallel)	57.5	volts rms	
Control phase, range (parallel)	0 to 63.25	volts rms	
Current:			
Reference phase	.114	amperes rms	maximum
Control phase (series)	.114	amperes rms	maximum
Control phase (parallel)	.210	amperes rms	nominal
Power:			
Reference phase	6.2	watts	maximum
Control phase (series)	6.2	watts	maximum
Control phase (parallel)	5.8	watts	nominal
Synchronous Speed	6000	rpm	
Impedance:			
Reference phase	514+j967	ohms	nominal
Control phase (series)	514+j967	ohms	nominal
Power Factor:			
Reference phase	.47		nominal
Control phase (series)	.47		nominal
Effective Resistance:			
Reference winding	2340	ohms	nominal
Control phase (series)	2340	ohms	nominal
Control phase (parallel)	585	ohms	nominal
Minimum Starting Voltage	3	volts rms	maximum
Coupling in Control Winding	.7	ratio	minimum
Mechanical:			
No-load Speed	5000	rpm	minimum
Stall Torque			
at 23°C	1.35	ounce-inches	minimum
at 125°C	1.05	ounce-inches	minimum

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TABLE I (Continued)

Requirements	Value	Unit	Tolerance
Rotor Moment of Inertia	3.3	gm-cm^2	maximum
Radial Play	.0008	inch	maximum
End Play	.0045	inch	maximum
	.0020	inch	minimum
Temperature Rise	85	degrees C	maximum
Noise, Mechanical	not excessive		
Time Constant	.0182	seconds	nominal
Reversing Time	.039	seconds	nominal
Theoretical Acceleration at Stall	29000	radians/second^2	nominal
Maximum Power Output	1.18	watts	nominal
Weight	8	ounces	maximum

Notes:

1. The servomotor identified herein shall conform to all requirements, and shall be subjected to all tests delineated in MIL-S-22432, with the following exceptions:

Salt Atmosphere Resistance, paragraph 3.30
Audible Noise, paragraph 3.31

2. Type Mk 7 Mod 7 servomotors shall be capable of operation in ambient temperatures ranging from -55°C to 125°C.

3. Connection Diagram:

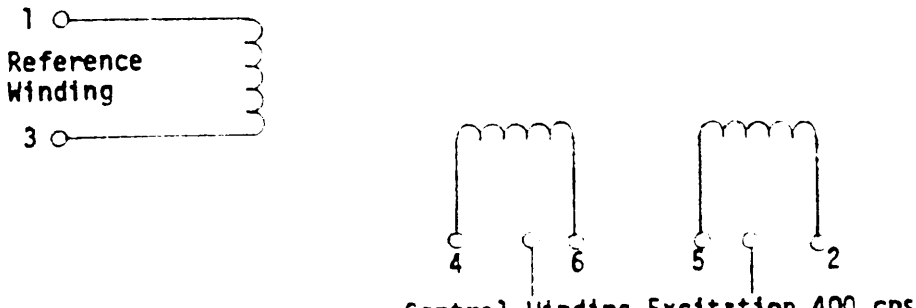


Figure 2

a. For CCW rotation the voltage at terminal 2 leads the voltage at terminal 1 by 90 electrical degrees.
b. For series operation of control winding terminals 5 and 6 are strapped and 115 volts rated is applied across terminals 2 and 4.
c. For multiple or parallel operation of control winding terminals 6 and 2 are strapped, and terminals 4 and 5 are strapped. 57.5 volts rated is applied across terminals 2 and 4.

4. Sampling, inspection and preparation for delivery shall be in accordance with MIL-S-22432.

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the minute after the oil appears through the transparent plastic line

MIL-S-22432/6B

Custodians:

Army - MU
Navy - AS
Air Force - 11

Review Activities:

Army - EL, MI, MU
Navy - SH, AS, OS, EC
Air Force - 11, 82

User Activities:

Army - ME, AV
Navy - OS, MC, CG

Preparing Activity:

Navy - AS

DOD Project Number:

6105-0107

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

MIL-S-22432/6B SERVOMOTOR, TYPE MK 7 MOD 7

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

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